

Harmonic Run-and-Tumble Dynamics: Beta Stationarity, Correlations, and Polynomial Jordan Resonances

Route-A source-local certificate HCS-C328

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Abstract

We close the invariant law of a one-dimensional harmonic run-and-tumble particle, including its two orientation components and every polynomial moment. The full stationary two-by-two correlation matrix is explicit, with its coalescent Jordan formula at $\mu = 2\lambda$.

1 Stationary beta law and all moments

Let $\sigma_t \in \{-1, 1\}$ flip to its opposite at rate λ , and let

$$dx_t = (v\sigma_t - \mu x_t) dt, \quad \mu, v, \lambda > 0. \quad (1)$$

The generator is

$$Lf_\sigma = (v\sigma - \mu x)\partial_x f_\sigma + \lambda(f_{-\sigma} - f_\sigma).$$

Set $a = v/\mu$, $y = x/a$, and $\alpha = \lambda/\mu$.

Theorem 1 (Stationary components and moments). *The process has a unique stationary probability on $[-a, a] \times \{-1, 1\}$. With respect to dy , its marginal and joint component densities are*

$$p(y) = \frac{\Gamma(\alpha + \frac{1}{2})}{\sqrt{\pi} \Gamma(\alpha)} (1 - y^2)^{\alpha-1}, \quad (2)$$

$$p_+(y) = \frac{1+y}{2} p(y), \quad p_-(y) = \frac{1-y}{2} p(y). \quad (3)$$

For every $n \geq 0$,

$$\mathbb{E}x^{2n} = a^{2n} \frac{(1/2)_n}{(\alpha + 1/2)_n}, \quad \mathbb{E}x^{2n+1} = 0, \quad (4)$$

$$\mathbb{E}(\sigma x^{2n}) = 0, \quad \mathbb{E}(\sigma x^{2n+1}) = a^{2n+1} \frac{(1/2)_{n+1}}{(\alpha + 1/2)_{n+1}}. \quad (5)$$

Proof. The deterministic vector fields point inward at $\pm a$. In stationarity, adding the two forward equations makes total probability flux constant; the no-flux boundary gives $(1-y)p_+ - (1+y)p_- = 0$. Hence (3) holds. Substitution in either remaining equation gives $p'/p = -2(\alpha-1)y/(1-y^2)$, and beta normalization gives (2). Positive flip rate makes the compact process irreducible in its accessible interior, giving uniqueness. Symmetry gives odd moments zero. Finally, integration by parts applied to $y^{2n+1}(1-y^2)^\alpha$ yields

$$\frac{\mathbb{E}y^{2n+2}}{\mathbb{E}y^{2n}} = \frac{n+1/2}{n+\alpha+1/2},$$

The component imbalance in (3) is $yp(y)$, so parity gives every even mixed moment zero and the displayed odd mixed moments. $\square$

2 Complete stationary correlation matrix

For $t \geq 0$, put $Z = (x, \sigma)^\top$ and $R(t) = \mathbb{E}[Z_t Z_0^\top]$. Since $Lx = -\mu x + v\sigma$ and $L\sigma = -2\lambda\sigma$,

$$R(t) := \mathbb{E}[Z_t Z_0^\top] = e^{At}\Sigma, \quad A = \begin{pmatrix} -\mu & v \\ 0 & -2\lambda \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \frac{v^2}{\mu(\mu+2\lambda)} & \frac{v}{\mu+2\lambda} \\ \frac{v}{\mu+2\lambda} & 1 \end{pmatrix}. \quad (6)$$

For $\mu \neq 2\lambda$, the upper-right entry of e^{At} is

$$b(t) = v \frac{e^{-2\lambda t} - e^{-\mu t}}{\mu - 2\lambda}.$$

Thus all four entries are fixed by (6); explicitly,

$$R_{xx}(t) = \frac{v^2[\mu e^{-2\lambda t} - 2\lambda e^{-\mu t}]}{\mu(\mu+2\lambda)(\mu-2\lambda)}, \quad (7)$$

$$R_{x\sigma}(t) = e^{-\mu t} \frac{v}{\mu+2\lambda} + b(t),$$

$$R_{\sigma x}(t) = e^{-2\lambda t} \frac{v}{\mu+2\lambda}, \quad R_{\sigma\sigma}(t) = e^{-2\lambda t}. \quad (8)$$

On the Jordan face $\mu = 2\lambda$, $b(t) = vte^{-\mu t}$ and

$$R_{xx}(t) = \frac{v^2(1 + \mu t)e^{-\mu t}}{2\mu^2}. \quad (9)$$

These are removable limits, not an omitted parameter case. For negative lag, stationarity gives the separate convention $R(-t) = R(t)^\top$ for $t \geq 0$; the one-sided semigroup formula in (6) is not applied with negative time.

At $\mu = 0$ there is no compact stationary probability on the line. At $\lambda = 0$, arbitrary mixtures of the two attracting endpoint atoms are stationary. At $v = 0$ with $\lambda > 0$, the joint stationary law is δ_0 tensored with uniform orientations; the polynomial couplings below disappear and all repeated diagonal values are semisimple. At the intersection $v = \lambda = 0$, orientation is frozen and every orientation mixture over δ_0 is stationary. For positive parameters the density in (2) diverges, is uniform, or vanishes at the endpoints according as $\alpha < 1$, $\alpha = 1$, or $\alpha > 1$; it has no endpoint atoms.

Source lineage

References

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- [2] R. Garcia-Millan and G. Pruessner, “Run-and-tumble motion in a harmonic potential: field theory and entropy production,” *J. Stat. Mech.* (2021), 063203. DOI: [10.1088/1742-5468/ac014d](https://doi.org/10.1088/1742-5468/ac014d).